LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 3 Test (GSEB pattern): Metals and Non-metals

Chapter / Topic: Metals and Non-metals

For teacher use only. Do not distribute to students.

SECTION A

Q1. Which metal is a liquid at room temperature? **[1]**

Answer: (a) Mercury. Bromine is a liquid too, but it is a non-metal.

Q2. Which non-metal is a liquid at room temperature? **[1]**

Answer: (d) Bromine (mercury is a liquid metal)

Q3. Which of the following can be beaten into thin sheets? **[1]**

Answer: (b) Zinc. Metals are malleable.

Q4. The property of metals that allows them to be drawn into thin wires is: **[1]**

Answer: (c) ductility

Q5. Which metal is stored under kerosene? **[1]**

Answer: (c) Sodium. It reacts so vigorously with air and moisture that it can catch fire.

Q6. Which of these metals reacts only with steam, not with cold or hot water? **[1]**

Answer: (b) Iron

Q7. A metal oxide that reacts with both acids and bases to form salt and water is called: **[1]**

Answer: (d) an amphoteric oxide, e.g. Al_2O_3, ZnO

Q8. In electrolytic refining, the anode mud consists of: **[1]**

Answer: (a) insoluble impurities

Q9. Ionic compounds conduct electricity: **[1]**

Answer: (c). The ions are free to move only in the molten state or in solution.

SECTION B

Q10. Zinc gives hydrogen gas with dilute hydrochloric acid, but copper does not. Why? Write the balanced equation for the reaction of zinc. **[2]**

Answer: Copper is less reactive than hydrogen (it is below hydrogen in the reactivity series), so it cannot displace hydrogen from the acid. Zinc is more reactive than hydrogen: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

- Reason (reactivity series): 1
- Balanced equation: 1

Q11. Why is sodium kept immersed in kerosene oil? **[2]**

Answer: Sodium reacts so vigorously with oxygen and moisture in the air that it catches fire if kept in the open. Kerosene keeps air and moisture away from it, preventing accidental fires.

- Very reactive with air/moisture, catches fire: 1
- Kerosene keeps air and water away: 1

Q12. What are amphoteric oxides? Give two examples. **[2]**

Answer: Metal oxides that react with both acids and bases to give salt and water. Examples: aluminium oxide (Al_2O_3) and zinc oxide (ZnO). For example, $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ and $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$.

- Definition: 1
- Two examples: ½ + ½

SECTION C

Q13. Answer the following about the metals aluminium, copper, iron and zinc: **[3]**

(i) Al > Zn > Fe > Cu [1]

(ii) No. Copper is less reactive than iron, so it cannot displace iron. [1]

(iii) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ [1]

Q14. Element X has atomic number 12, and element Y has atomic number 17. **[3]**

(i) X: 2, 8, 2 (magnesium). Y: 2, 8, 7 (chlorine). [1]

(ii) XY_2 ($MgCl_2$). X loses 2 electrons and each of two Y atoms gains 1 electron. [1]

(iii) Ionic bond. Properties (any two): high melting point; soluble in water; conducts electricity when molten or in solution. [1]

SECTION D

Q15. Pure gold, known as 24 carat gold, is very soft and is not suitable for making jewellery. It is alloyed with silver or copper to make it hard. In… **[4]**

(i) 22 parts of pure gold alloyed with 2 parts of copper or silver. [1]

(ii) Copper and tin. [1]

(iii) Solder has a low melting point. Other reasons (any one): to increase hardness or strength, to reduce corrosion, to change electrical conductivity. [2]

- Low melting point: 1
- Another reason: 1